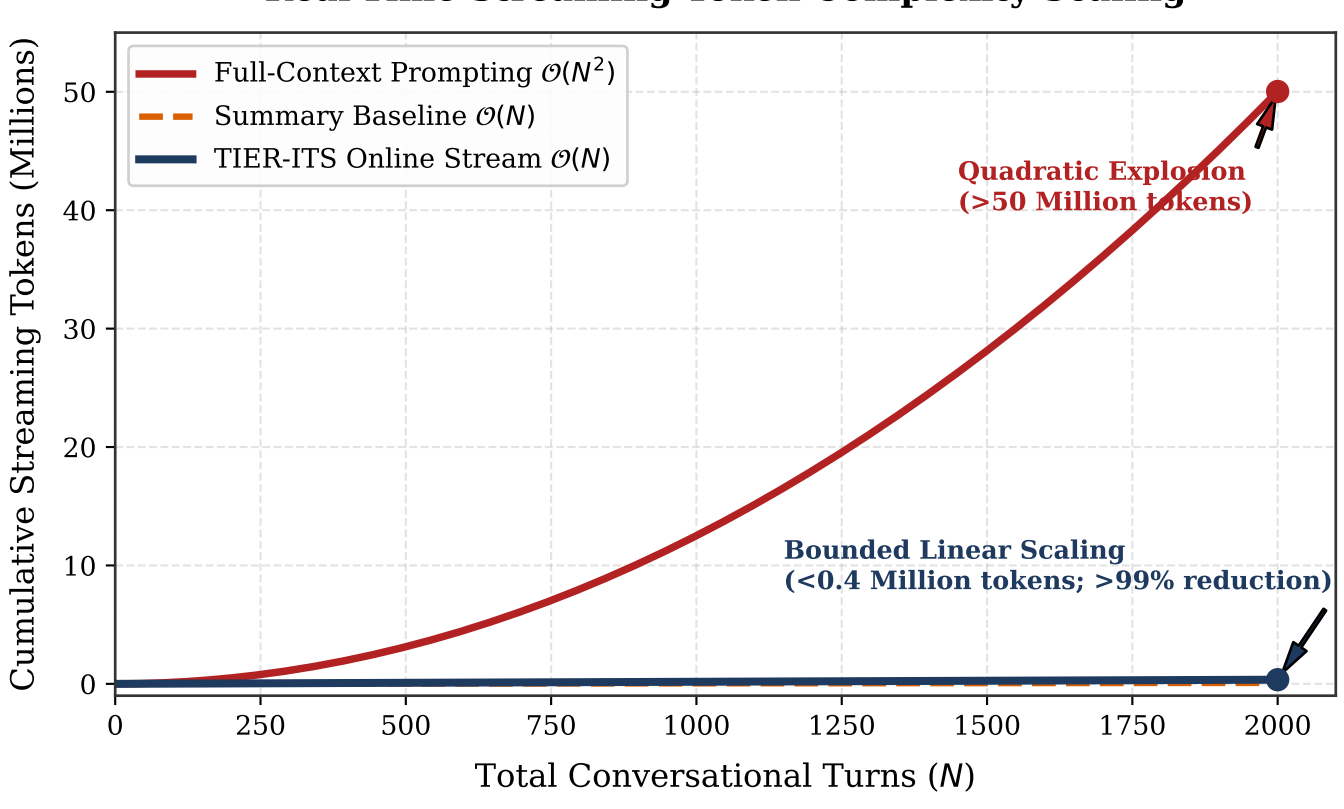
Real-Time Streaming Token Complexity Scaling
Full-Context Prompting $\mathcal{O}(N^2)$
Summary Baseline $\mathcal{O}(N)$
TIER-ITS Online Stream $\mathcal{O}(N)$
Quadratic Explosion
(>50 Million tokens)
Bounded Linear Scaling
(<0.4 Million tokens; >99% reduction)
Cumulative Streaming Tokens (Millions)
0
10
20
30
40
50
0
250
500
750
1000
1250
1500
1750
2000
Total Conversational Turns (N)